

THE ANATOMY OF AN LSTM CELL

Mastering Long-Term Memory through Vector Traffic Control.

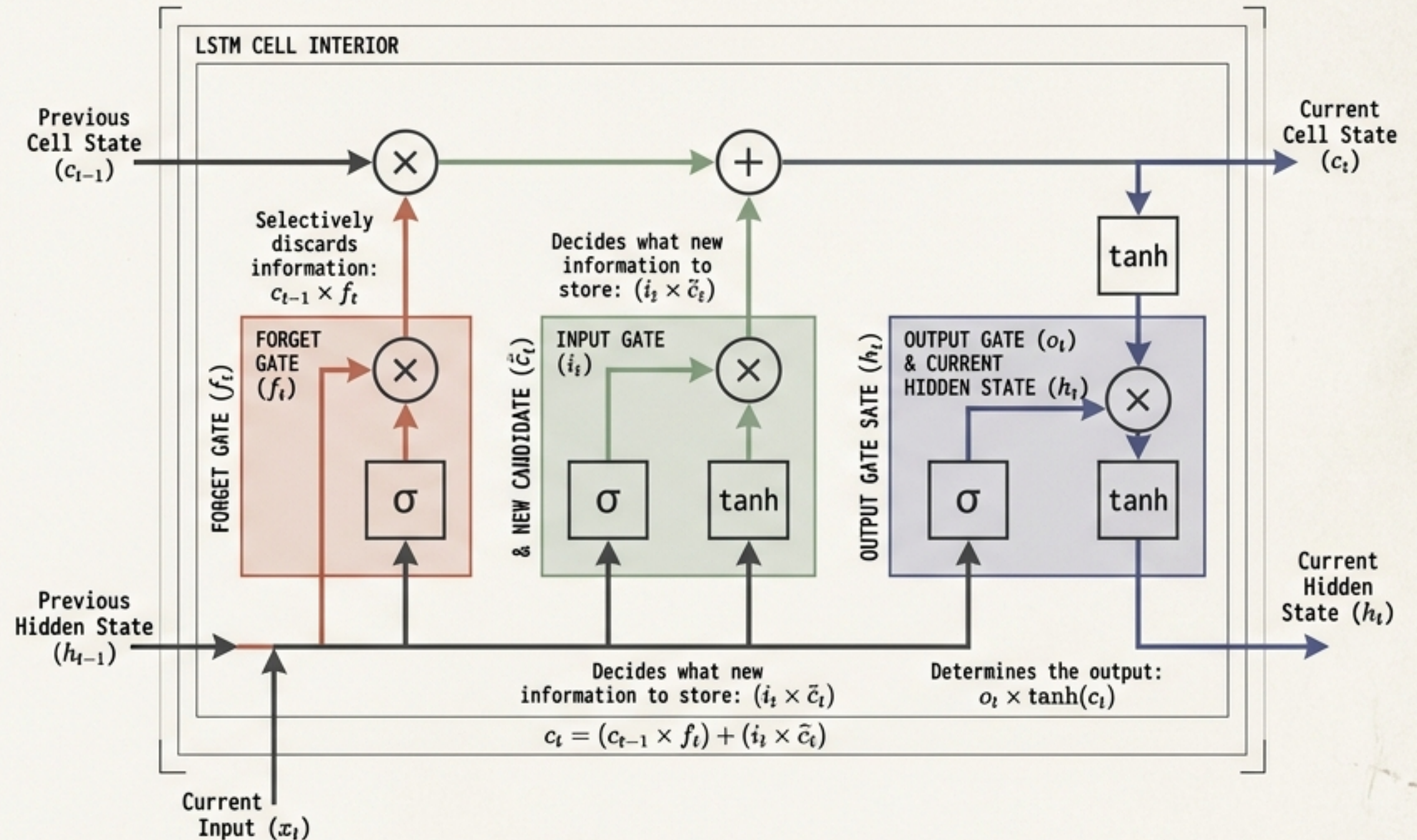
An LSTM (Long Short-Term Memory) cell operates as a sophisticated traffic control system for information flow, utilizing specialized gates to regulate the storage, retrieval, and discarding of data over extended sequences. The cell state, ' c_t ', acts as a conveyor belt, with linear interactions governed by three primary gates: the Forget Gate, Input Gate, and Output Gate, each functioning through sigmoid (' σ ') and hyperbolic tangent (' $\tanh$ ') activations, represented mathematically as:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f),$$

$$j_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i),$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \text{ and}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o).$$

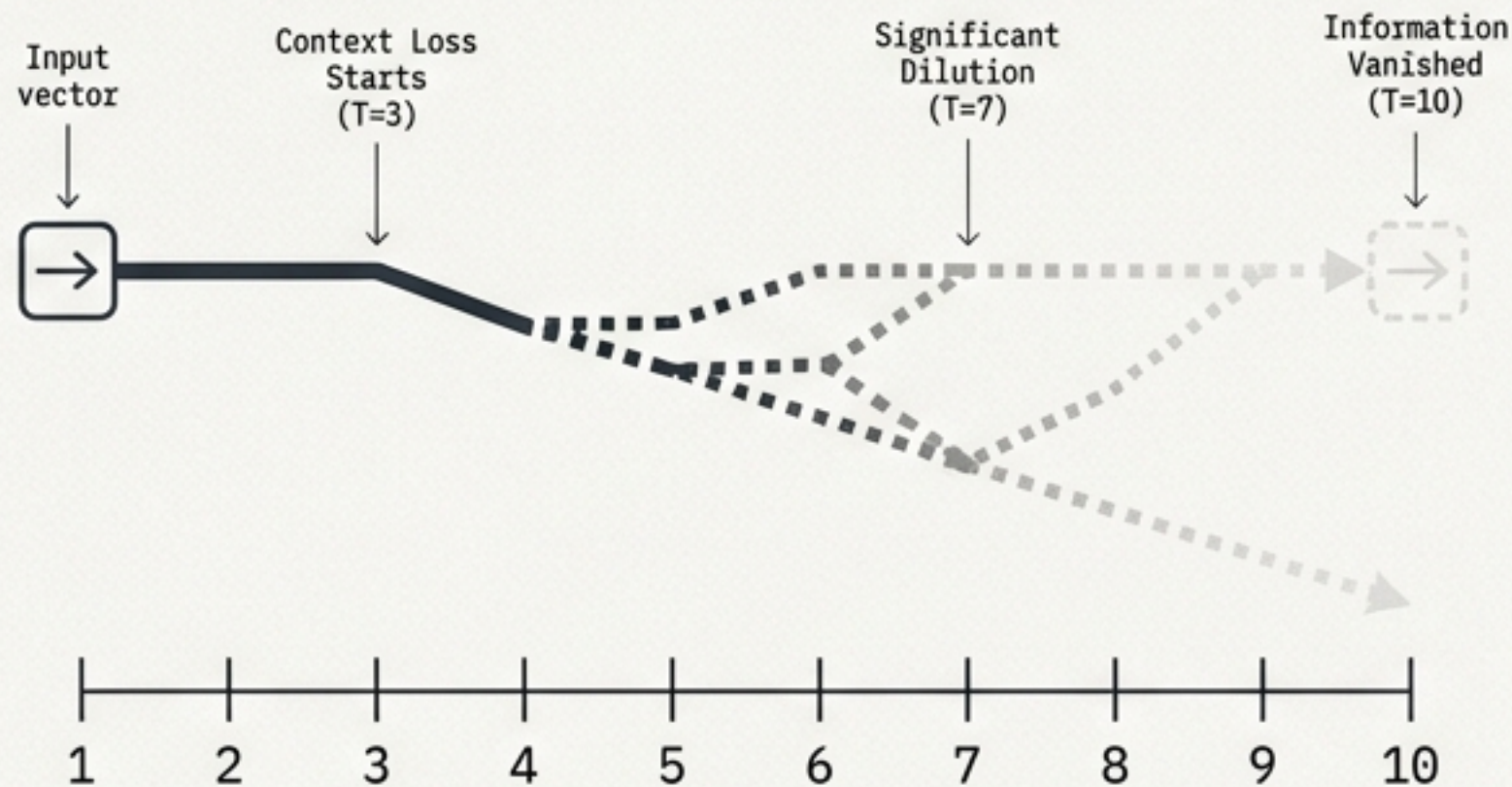


THE VANISHING CONTEXT PROBLEM

Simple RNNs dilute long-term context over time. LSTMs preserve exact vectors across massive sequence lengths using mathematical traffic lights.

SHORT-TERM MEMORY (STM) / VANISHING GRADIENT

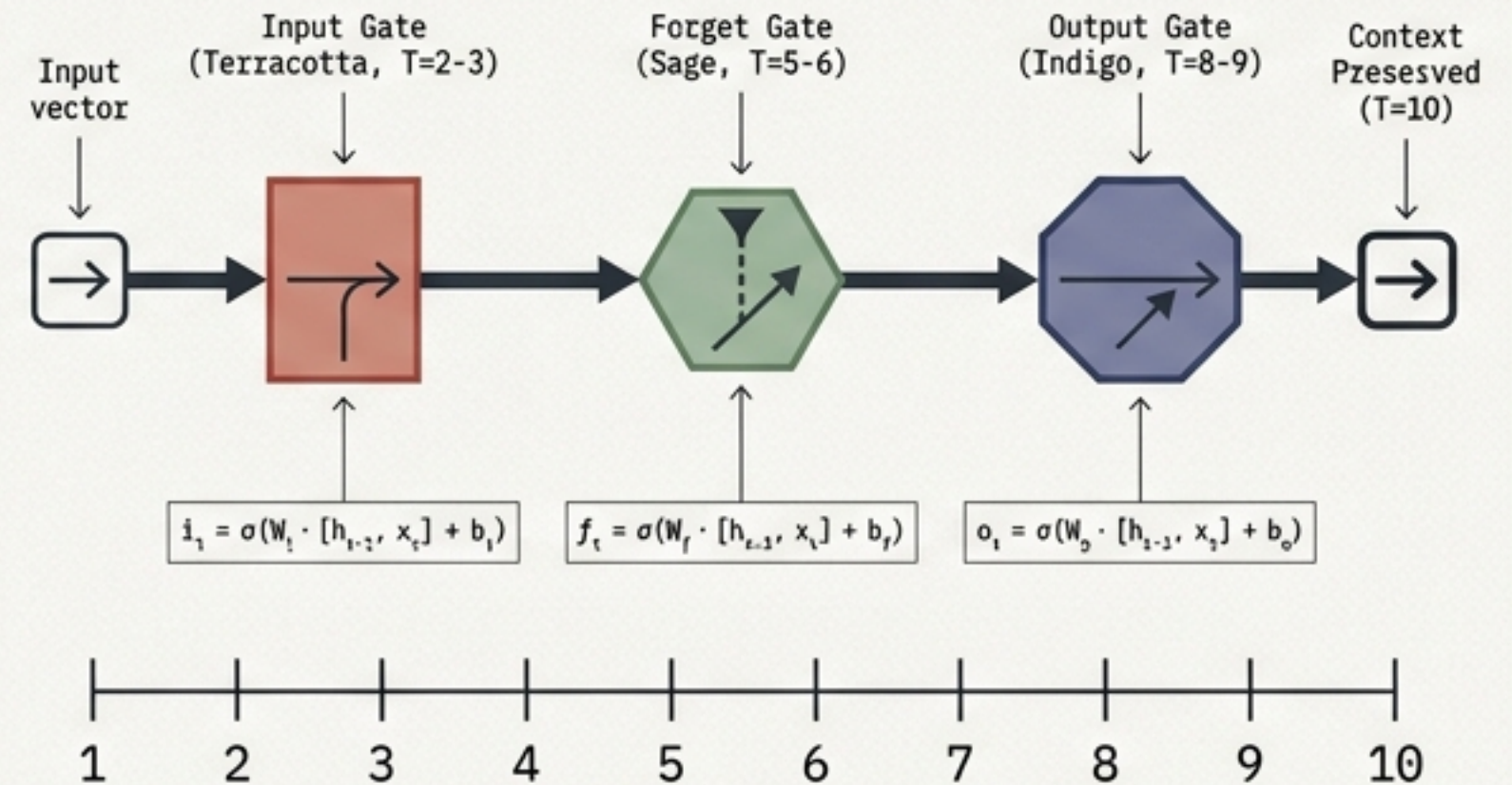
■ TERRACOTTA: WARNING/DEGRADATION



GRADIENT FLUX ≈ 0 AT T=10

LONG-TERM MEMORY (LTM) / CONTROLLED VECTOR TRANSFER

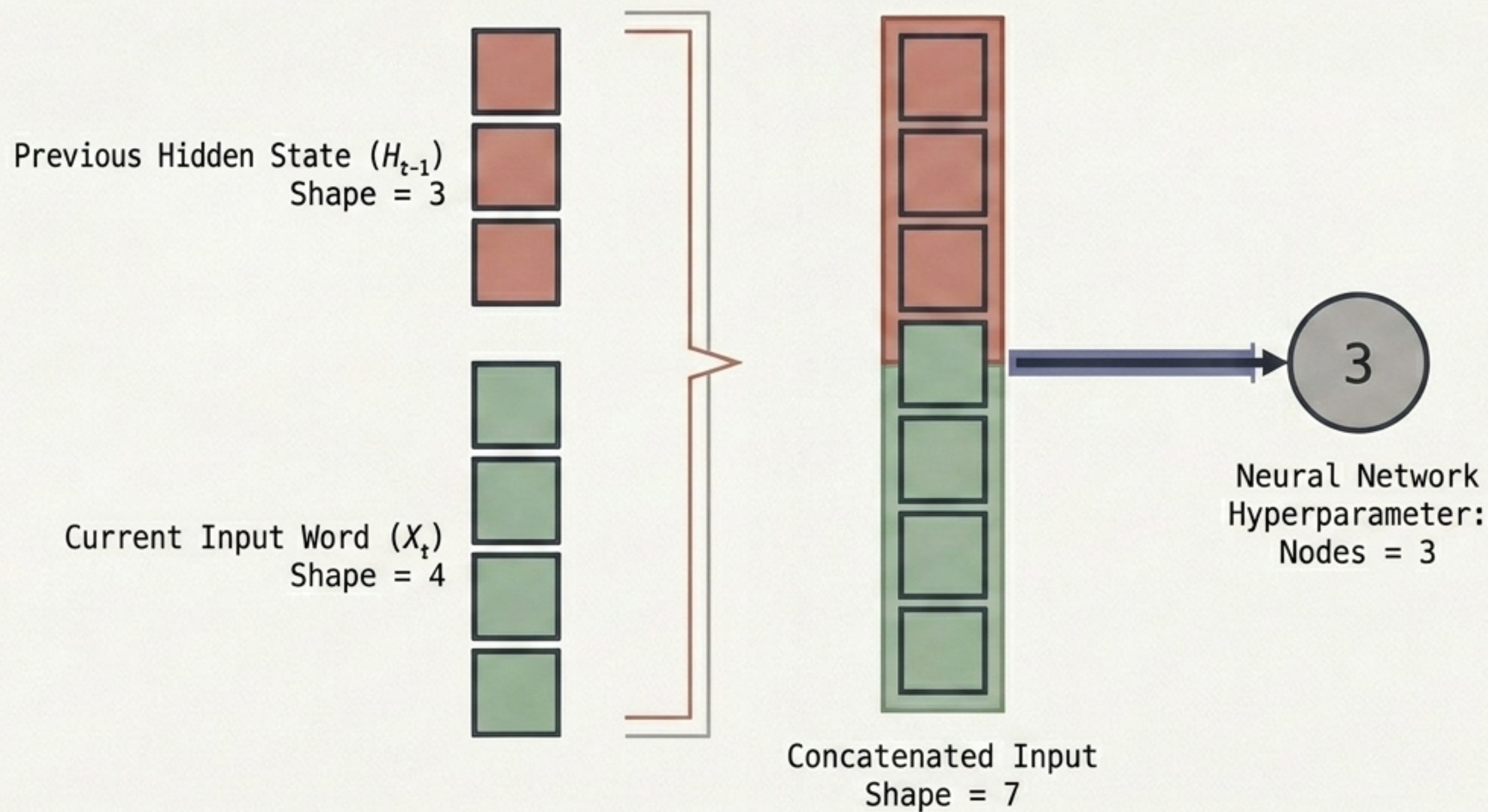
■ TERRACOTTA: INPUT/GATE CONTROL
■ SAGE: FORGET/MEMORY CONTROL
■ INDIGO: OUTPUT/STATE TRANSFER



CELL STATE VECTOR PRESERVED UNCHANGED

ESTABLISHING THE CORE DIMENSIONS

Because our hidden size is 3, all memory states must be exactly size 3.



GATE 1: THE FORGET GATE

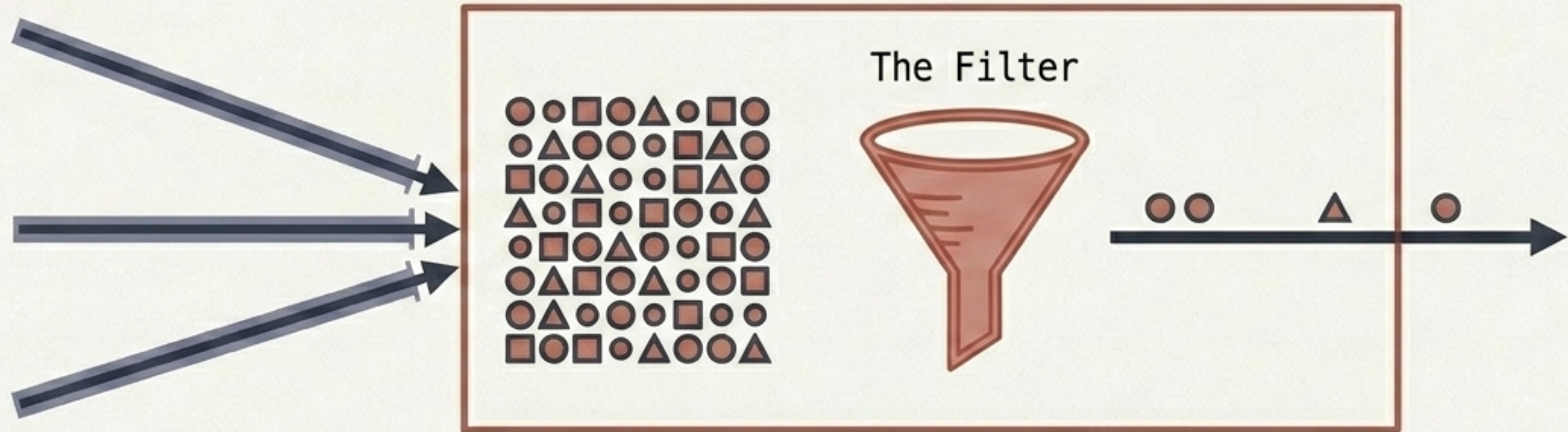
When new context arrives, old information clutters the memory.
The Forget Gate calculates exactly what to erase.

Inputs:

C_{t-1} (Long-Term Memory)

h_{t-1} (Short-Term Memory)

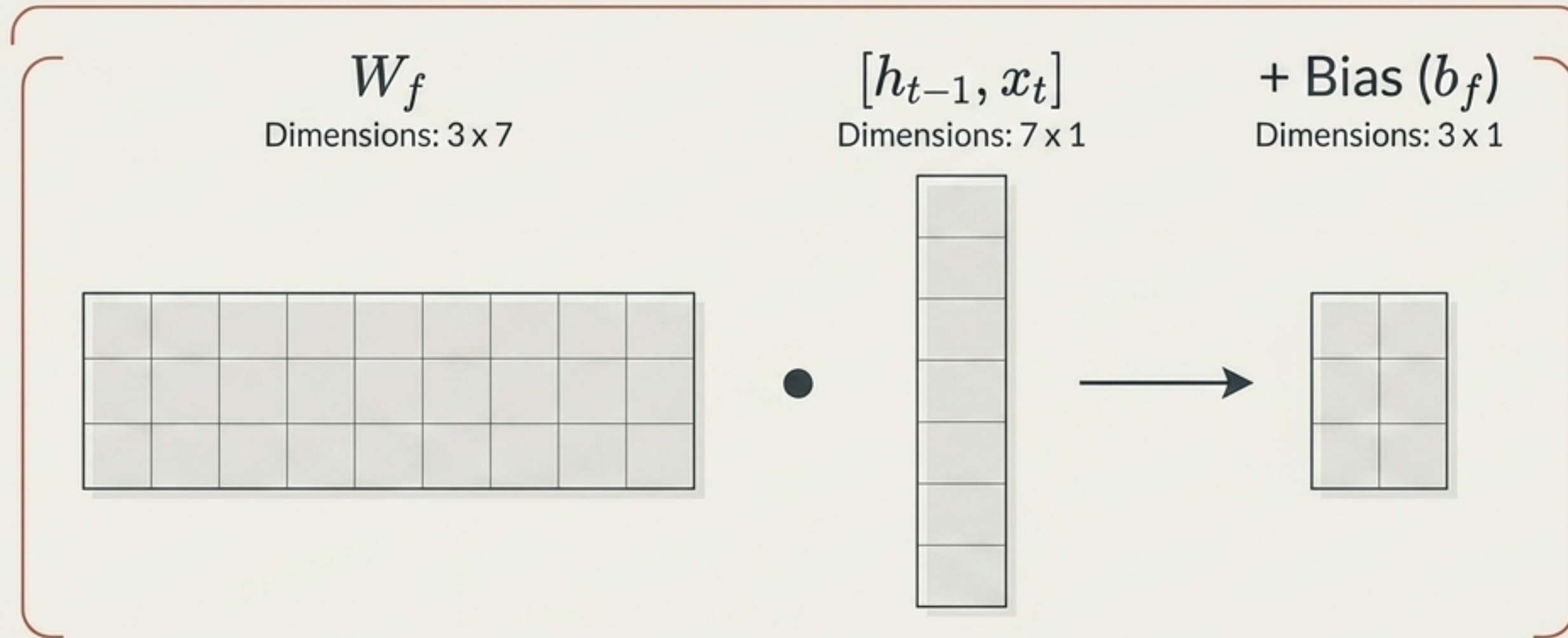
x_t (Current Input)



GENERATING THE FORGET VECTOR (f_t)

The 3x7 weight matrix maps 7 inputs perfectly to our 3 required memory nodes.

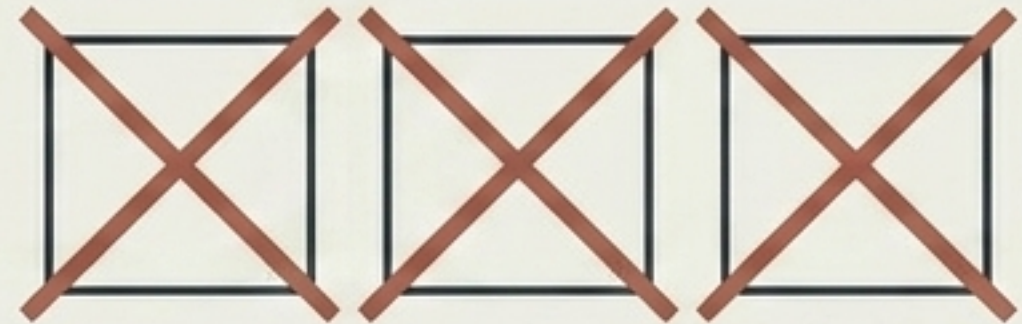
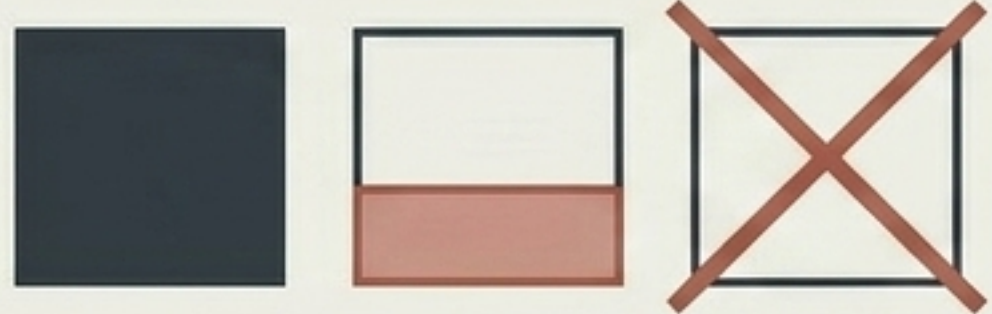
Sigmoid (σ) Activation



$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

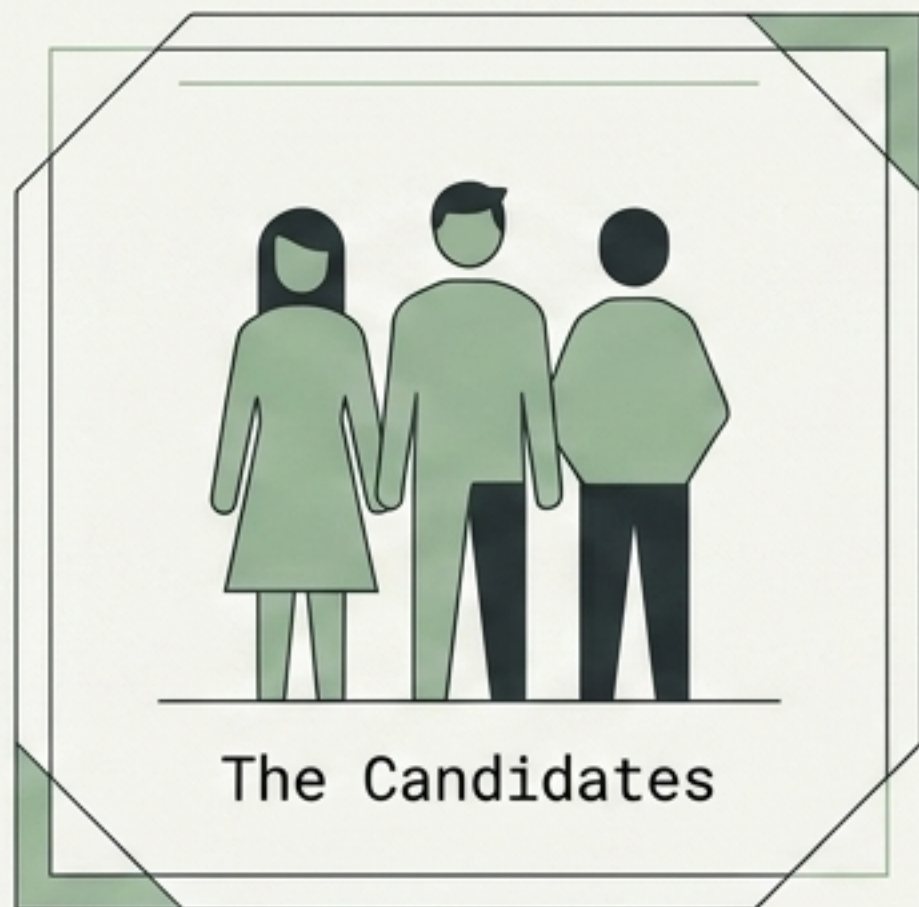
Pointwise Application: $C_{t-1} \otimes f_t$

The Sigmoid function (0 to 1 range) enables absolute precision over what is kept, scaled, or destroyed.

Case 1: 100% Retained	Old Memory [1.2, 3.1, 2.1] $\otimes$ Forget Vector [1, 1, 1] = Result [1.2, 3.1, 2.1]	
Case 2: 0% Retained (Wiped)	Old Memory [1.2, 3.1, 2.1] $\otimes$ Forget Vector [0, 0, 0] = Result [0, 0, 0]	
Case 3: Partial Forgetting (Scaled)	Old Memory [1.2, 3.1, 2.1] $\otimes$ Forget Vector [1, 0.2, 0] = Result [1.2, 0.62, 0]	

Gate 2: The Input Gate (Adding New Context)

Storing new information requires two steps: generating candidates, and filtering them.



Candidate Cell State ($\tilde{C}_t$)

The pool of potential new information.

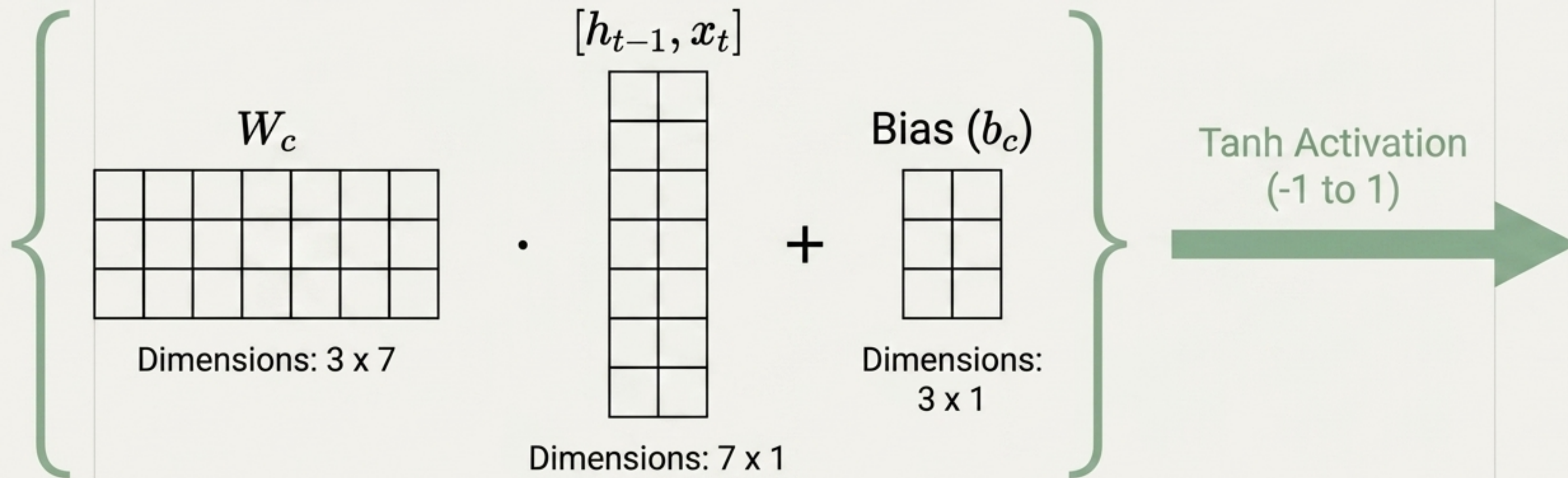


Input Vector (i_t)

The filter deciding which candidates are worthy.

Step 1: Generating the Candidates ($\tilde{C}_t$)

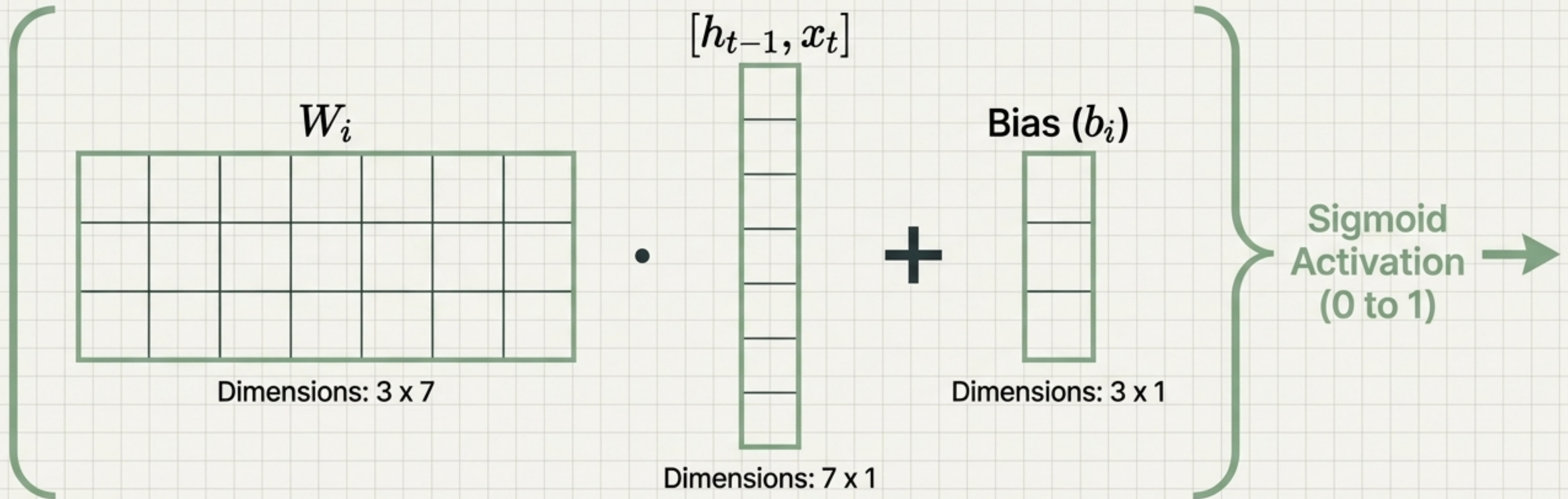
Tanh normalizes all potential new memory values to a standardized scale.



$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

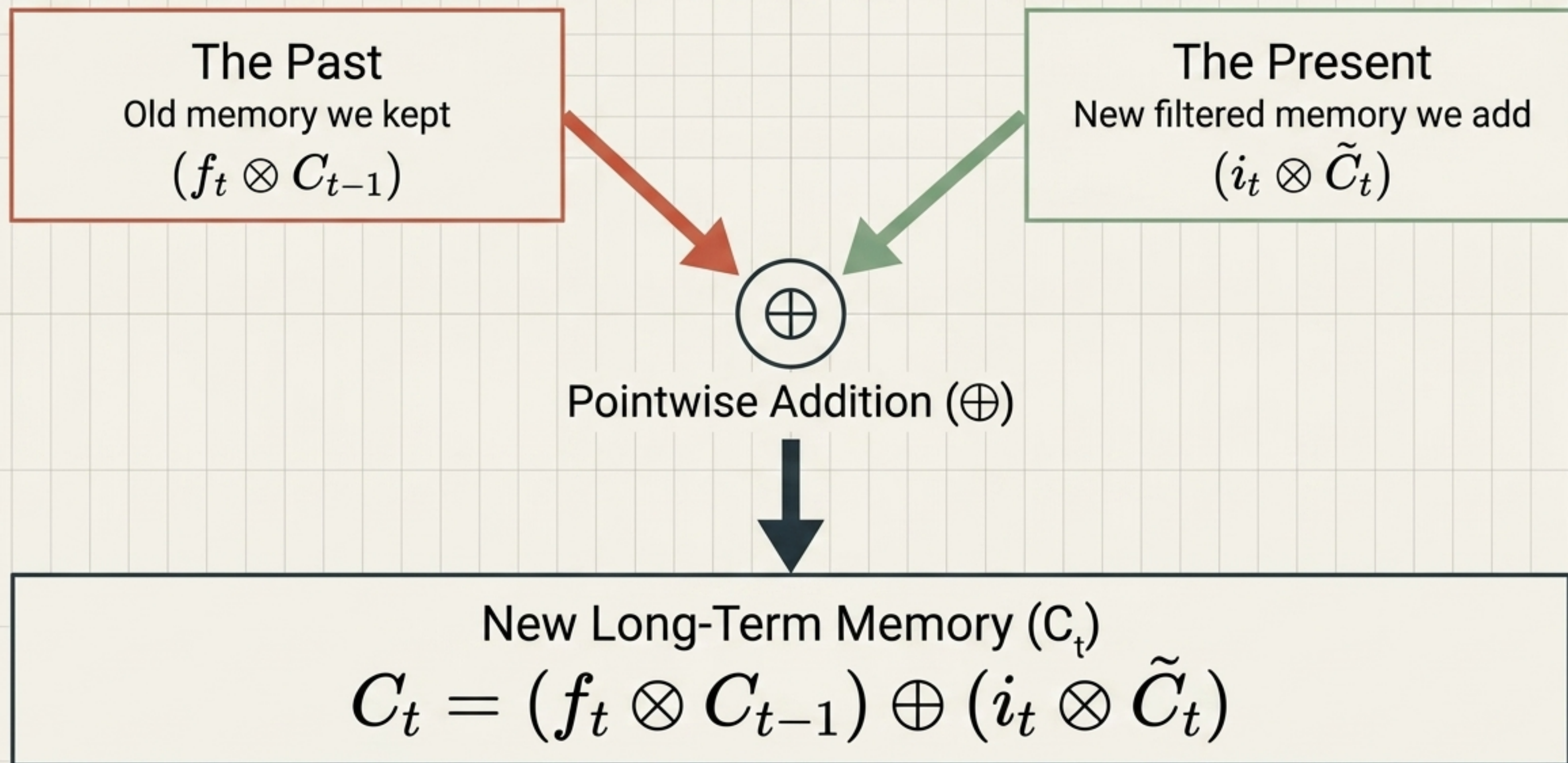
Step 2: Generating the Input Filter (i_t)

The **Sigmoid filter** decides how much of each candidate (from 0% to 100%) enters the system.



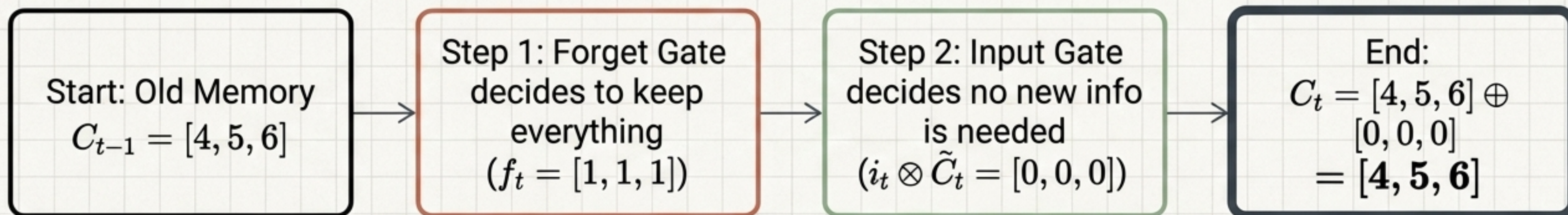
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

The Master Equation: Updating Long-Term Memory



The Core Magic of LSTM: Pure Vector Transfer

By simply outputting 1s (keep) and 0s (block), the original vector travels to the next time step without any loss or dilution.



Gate 3: The Output Gate

Clearing the Misconception: The Output Gate does NOT calculate C_t . It only uses the newly updated C_t to figure out the Short-Term Memory (h_t).

Long-Term Memory (C_t)

Flows directly to the next cell state.



Output
Filter

Short-Term Memory (h_t)

Current time step output.

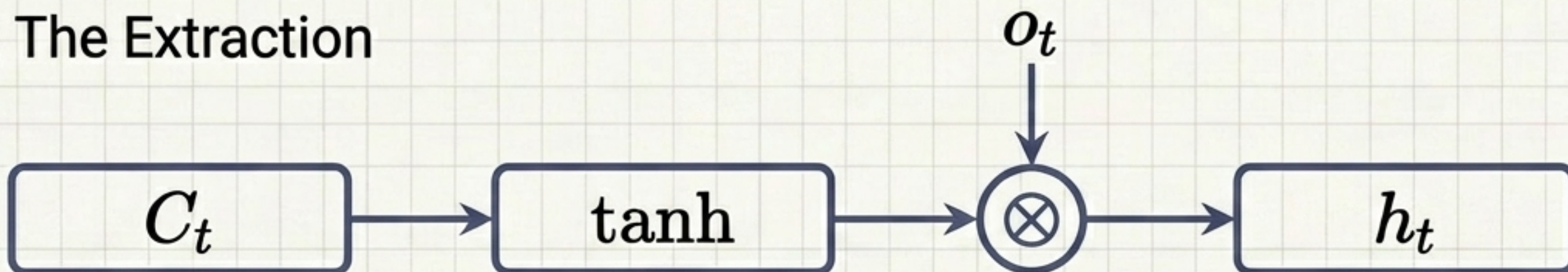
Calculating the Hidden State (h_t)

We scale the massive Long-Term Memory, then filter it to extract only what is immediately useful for the current time step.

Step 1: The Filter

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

Step 2: The Extraction



$$h_t = o_t \otimes \tanh(C_t)$$

The 3-Gate Mathematical Reference Guide

Forget Gate	Input Gate	Output Gate
Objective: Erase	Objective: Add	Objective: Extract
Activation: Sigmoid	Activation: Tanh & Sigmoid	Activation: Sigmoid & Tanh
Output: f_t	Output: $i_t, \tilde{C}_t$	Output: h_t
Math: $\sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$	Math: $\tanh(W_c \cdot ...) \times \sigma(W_i \cdot ...)$	Math: $o_t \otimes \tanh(C_t)$

Master Update: $C_t = (f_t \otimes C_{t-1}) \oplus (i_t \otimes \tilde{C}_t)$

The Ultimate Vector Traffic Controller

At its core, an LSTM is a highly controlled Vector Transfer System. By acting as traffic lights—calculating whether to transfer, block, or modify vectors—gates achieve pure long-term memory at scale.

